

Program : Diploma in Chemical Engineering	
Course Code : 5079	Course Title: Environmental Engineering Lab II
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To enable the students to apply the water treatment methods for drinking water and effluents.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Environmental Science		Environmental Science	2

Course Outcomes :

On completion of the course student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Practice the removal of hardness from water	12	Applying
CO2	Practice the treatment of effluent water	21	Applying
CO3	Practice heavy metal removal from effluent water.	9	Applying
CO4	Practice colour removal from effluent water.	3	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3				3		
CO3	3				3		
CO4	3				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Practice the removal of hardness from water		
M1.01	Remove temporary hardness from water.	3	Applying
M1.02	Estimate the removal efficiency of temporary hardness.	3	Applying
M1.03	Remove permanent hardness from water.	3	Applying
M1.04	Estimate the removal efficiency of permanent hardness.	3	Applying
CO2	Practice the treatment of effluent water		
M2.01	Estimate the amount of dissolved oxygen in given effluent sample.	3	Applying
M2.02	Estimate the COD of given effluent sample.	3	Applying
M2.03	Estimate the BOD of given effluent sample.	3	Applying
M2.04	Remove the COD and BOD of given effluent sample by. chemical precipitation.	3	Applying
M2.05	Estimate the percentage removal of COD	3	Applying
M2.06	Estimate the percentage removal of BOD	3	Applying
M2.07	Open ended experiments	3	Applying
	Series Test – I		
CO3	Practice heavy metal removal from effluent water.		
M3.01	Estimate the amount of hexavalent chromium in given sample.	3	Applying
M3.02	Remove hexavalent chromium from given sample by chemical precipitation.	3	Applying
M3.03	Estimate the percentage removal of chromium from given sample.	3	Applying
CO4	Practice colour removal from effluent water.		
M4.01	Estimate the colour and then remove the colour from given effluent water by adsorption.	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Standard Methods for the Examination of Water and Wastewater, 22nd Edition ; Eugene W. Rice, Rodger B. Baird, Andrew D. Eaton
R1	Wastewater engineering: treatment and reuse; Metcalf & Eddy
R2	Principles and Practices of Air Pollution Control and Analysis; J R Mudakavi

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/107/103107084/